Universität Zürich[UZH]

• Institute of Psychology

Modelling Recall Errors in Continous Reproduction Tasks

Satellite Event: Analyzing data on the level of cognitive processes

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ESCoP 2025

Workshop: Overview

Monday

Morning

Intro

Why hierarchical models?

Signal-Detection Models (brms)

Afternoon

Memory Measurement Model (bmm)

Practical Exercises

Tuesday

Morning

Diffusion Decision Model (bmm)

Practical Exercises

Afternoon

Signal Discrimination Model

Reporting Results from Cognitive Modelling Analyses

Materials are available via GitHub

30 Min. Coffee Breaks
10:30 & 15:00

60 Min. Lunch Break
12:30

Ask questions whenever something is not clear
☺

Traditional Models of VWM

Signal Discrimination Model

Two-Parameter Mixture Model

(Zhang & Luck, 2008)

P_{mem} | Resp. Error | Kappa

Three-Parameter Mixture Model

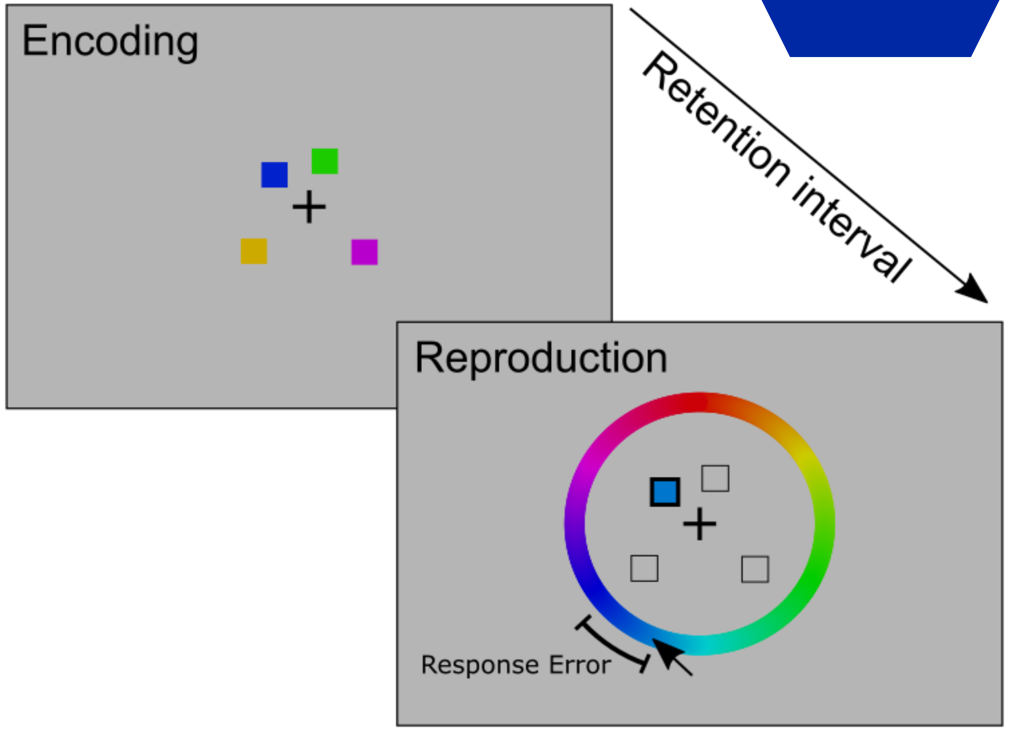
(Bays, 2009)

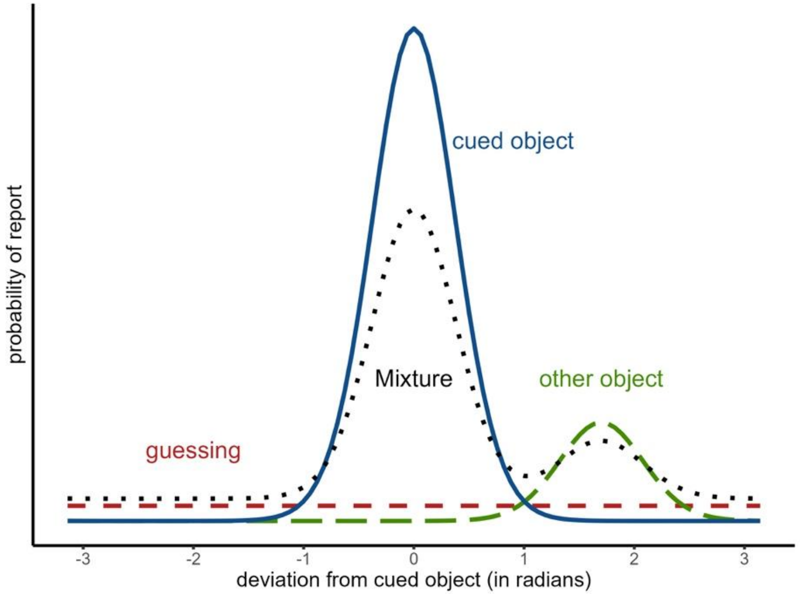
P_{mem} | P_{swap} | Kappa | Resp. Error

Interference Measurement Model

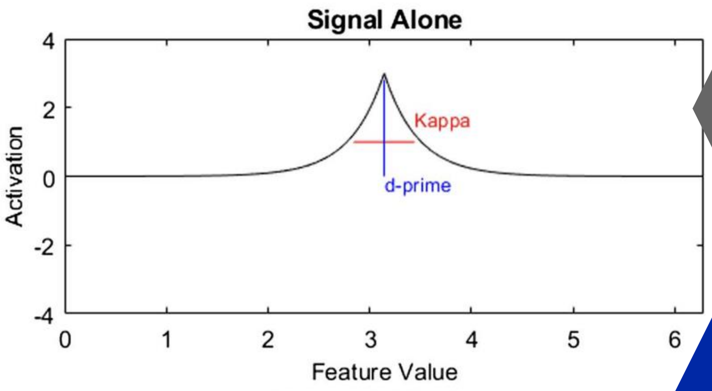
(Oberauer et al., 2017)

Context Act. | General Act. | Context Generali-zation | Resp. Error | Kappa





Signal Detection Based Models of VWM



LaPlace Activartion Function

Complex to estimate ☹

TCC

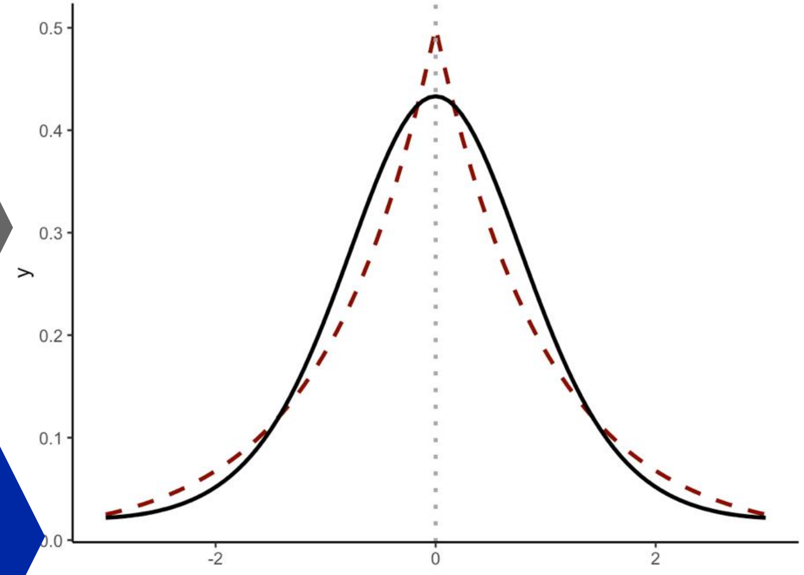
Gaussian Noise

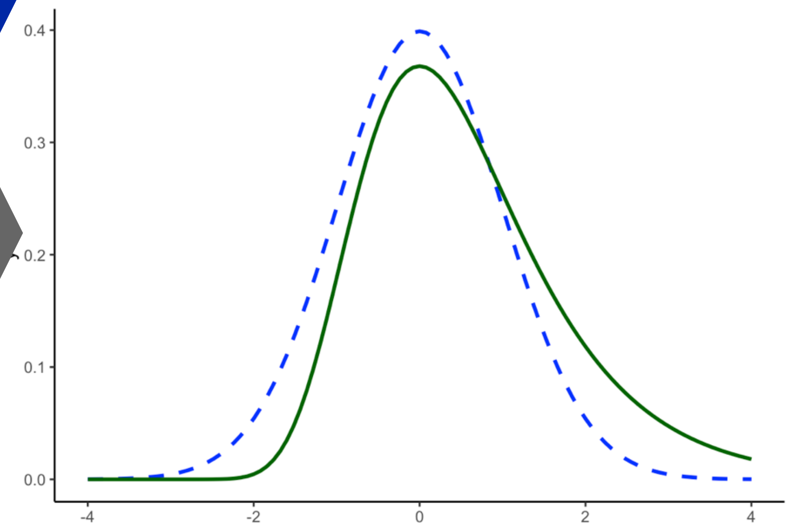
Simple to estimate ☺

VonMises Activation Function

SDM

Gumbel Noise





Feature Value

How to implement theSignal discrimination model in bmm?

ESDM– Model Object & Formula

Diffusion Decision Model (bmm)

- Again, we set up a *bmmodel* object for the EZ-Diffusion model using the `ezdm()` function:

```
sdm_model = sdm(
  resp_error
)
```

Response error in each trials of each condition

```
sdm_formula <- bmf(
  c ~ 1 + setsize + (1 + setsize | ID),
  kappa ~  1 + (1 | participant)
)
```

What are your questions so far?

References

- Bürkner, P. C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of statistical software*, *80*, 1-28.

- Carlisle, N. B. (2023). Negative and positive templates: Two forms of cued attentional control. *Attention, Perception, & Psychophysics*, *85*(3), 585-595.

- Frischkorn, G. T., & Popov, V. (2025). A tutorial for estimating Bayesian hierarchical mixture models for visual working memory tasks: Introducing the Bayesian Measurement Modeling (bmm) package for R. *Behavior Research Methods*, *57*(5), 144

- Tavares, G., Perona, P., & Rangel, A. (2017). The attentional drift diffusion model of simple perceptual decision-making. *Frontiers in neuroscience*, *11*, 468.